

Conformal Meta-learners for ITEs

Valid, model-agnostic predictive intervals

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Introduction

- Goal: predictive intervals for individual treatment effects (ITEs) with target coverage.
- Bayesian intervals are model-specific and may lack finite-sample frequentist coverage.
- Conformal prediction (CP) offers model-agnostic, distribution-free marginal coverage.

Key Challenges

- Covariate shift between treated/untreated vs. target population.
- Inductive biases from combining nuisance estimates to learn CATE.

Results: Findings & Observations

- DR delivers strong coverage and efficient intervals with low RMSE across datasets.
- Efficiency correlates with closeness of pseudo-outcome score CDF to oracle CDF.
- Large dominance gaps (e.g., in some NLSM settings) can widen intervals.

Combined summary: CDF dominance (DR/IPW), coverage vs. width (see Figures 4–5)

Main Takeaways

- Coverage: Conformal meta-learners achieve marginal validity for ITEs under stochastic dominance; DR/IPW often satisfy it.
- Efficiency: DR yields competitive, often shorter intervals while keeping low CATE RMSE.
- Dominance: Stronger dominance of pseudo-score CDFs over oracle can make intervals conservative.

Results: Summary

- IPW/DR often exhibit empirical first-order dominance over oracle scores $\Rightarrow$ broad validity.
- DR achieves strong coverage with comparatively short intervals and low CATE RMSE.
- X-learner attains low RMSE but under-covers due to lacking distribution-free order.

Summary visuals: CDF gaps, coverage vs. interval length (see paper Figures 4–5)

Abstract & Motivation

- Conformal meta-learners issue predictive intervals for ITEs atop CATE meta-learners.
- Use two-stage pseudo-outcome regression; calibrate with CP.
- Stochastic ordering links pseudo-outcome conformity scores to oracle scores for validity.
- DR learner satisfies model- and distribution-free dominance at relevant coverage levels.
- Directly infers ITEs and retains point-estimation strengths.

Problem Setup

- Binary treatment, covariates, and outcomes under potential outcomes framework.
- Observe factual outcome only; counterfactual is unobserved.
- Assume unconfoundedness, consistency, and positivity.
- Target: predictive band covering the true ITE on new individuals.

Conformal Prediction (CP): Recap

- Split data into train and calibration; fit a base model on training data.
- Compute conformity scores on calibration data to quantify residual surprise.
- Form intervals by expanding around predictions using a calibration quantile.
- Guarantee: finite-sample marginal coverage under exchangeability.

Oracle CP for ITEs: Infeasible

- Would calibrate using true ITEs, which are never observed.
- Necessitates proxy calibration using only observed variables.

Illustration: oracle conformity scores require unobserved counterfactuals

Naïve ITE Inference Pitfalls

- Separately conformalizing outcome models can break exchangeability via covariate shift.
- Ties inference to nuisance models and yields conservative ITE intervals.

Qualitative Example (Toy Case)

Case: Extreme Propensity

Toy scatter: as propensity $\rightarrow$ 0/1, pseudo-outcome variability increases $\rightarrow$ wider intervals

Case: Balanced Propensity

Toy scatter: near-balanced treatment assignment $\rightarrow$ tighter intervals

- Notice intervals widen when propensity is extreme.
- DR pseudo-outcomes reduce variance vs. pure IPW in these regions.
- Calibration uses the same covariate distribution as test, mitigating shift.

Conformal Meta-learners: Overview

- Two-stage pseudo-outcome regression plus post-hoc CP on pseudo-outcomes.
- Addresses covariate shift via calibration on target-distribution pseudo-outcomes.
- Decouples CP from model architecture; enables direct ITE inference.

Stage 1: Nuisance

Estimate outcome models (propensity known)

Stage 2: Pseudo-outcome

Build pseudo-outcomes from observed data and nuisance estimates

Calibration

Conformity on pseudo-outcomes $\Rightarrow$ interval for ITE

Conformal Pseudo-intervals

- Split into three subsets: nuisance estimation, CATE training, and calibration.
- Compute conformity scores on held-out pseudo-outcomes to get a calibration quantile.
- ITE interval formula: $\hat{C}\phi(x) = [\tau(x) \pm Q_{V\phi}(1-\alpha)]$.

Exchangeability holds for (X, pseudo-outcome) in calibration vs. test; need stochastic ordering to transfer coverage to true ITEs.

Validity & Learner Guarantees

Stochastic Ordering Framework

- Sufficient if pseudo-outcome scores dominate or suitably spread vs. oracle scores.
- Coverage holds above a threshold; full range when first-order dominance holds.

Who Satisfies It?

- X-learner: no model- and distribution-free order vs. oracle.
- IPW: convex dominance (absolute residual) and related spread conditions.
- DR: same guarantees as IPW; often empirically stronger (first-order dominance).

Notes & Limits

- Intervals may be conservative due to dominance.
- Requires known propensity for distribution-free guarantees; exact coverage threshold depends on data.

Meta-learners via Pseudo-outcomes

- X-learner: uses regression adjustment to impute counterfactuals.
- IPW-learner: reweights outcomes so pseudo-outcome matches CATE in expectation.
- DR-learner: combines IPW and regression adjustment; improved accuracy and efficiency.

Experiments: Setup

- Synthetic data with heteroscedastic noise; two setups (no effect vs. heterogeneous effects).
- Semi-synthetic: IHDP and NLSM with real covariates and simulated outcomes.
- Baselines: Weighted CP variants vs. conformal meta-learners (DR/IPW/X).
- Metrics: coverage, interval length, and CATE RMSE.